

Dfmea

Design FMEA (Qualitative)

Virtual Battery Factory · case t7_r1 · 2026-08-25

Design FMEA (Qualitative, Simulation-Signal Based) — Case t7_r1

Document: VBF-T7R1-DFMEA-01 · Prepared 2026-08-25 · Signature: Prepared _____ / Reviewed _____ / Approved _____

Qualitative version based on simulation signals (S/O three-level: High/Medium/Low; magnitude = deviation of simulated value from threshold).
Complete FMEA including process/supplier failures: N/A (beyond pure-simulation boundary).

#	Failure mode	Failure cause	Simulation signal (detectability basis)	Severity	Occurrence	Design mitigation
1	Lithium plating on negative electrode during 4C fast charge	Transport/solid-diffusion polarization at high rate (large graphite particles, low porosity, low electrolyte transport)	anode_potential_v min: baseline -0.1918 V (plates) → design +0.0476 V; isolation data: porosity dominant lever (-0.134 → -0.0419 V), h=200 worsened (-0.131 V)	High (Li metal → short-circuit/TR hazard)	Low (margin +0.048 V at 4C/45 °C with 1.5 µm graphite, porosity 0.36, σ 2.5 S/m, t 0.55)	Keep porosity ≥ 0.36 and particle radius ≤ 1.5 µm; electrolyte transport overrides as specced; do not fast-charge below 0 °C (not simulated); re-verify after any material change
2	Thermal runaway from nail penetration	Internal short with 10 W heat source exceeding local heat rejection	run-tr triggered flag: baseline triggered 322.2 s (contract h=10) → design not triggered, T_max 336.32 K; sensitivity: h=30 triggers at 666 s, h=40 passes 348.69 K	High	Low (design h = 50 W/(m²·K), 25 % above the 40 floor)	Cooling spec h ≥ 40 W/(m²·K) on 0.00531 m² surface as a hard floor; monitor thermal-interface degradation in pack integration
3	Cell overheating during 4C fast charge	Ohmic + polarization heating at 45 °C ambient	T_max_K = 330.13 K (56.98 °C) at 45 °C ambient (h = 50)	Medium	Low	No T_max threshold adjudicated in task; 330 K is far below SEI runaway-onset regimes; keep h = 50
4	Electrolyte oxidative decomposition at high voltage	HOMO level of solvent vs cathode potential window (4.2 V upper cut-off)	No molecular data (Stage 2 not entered; start_stage = 3; real_compute = false)	Medium	Not assessed (honest gap — no proxy/DFT run)	Upper cut-off kept at parameter-set 4.2 V; first-principles endorsement skipped per real_compute=false — flagged for future work
5	Insufficient capacity / energy density	Cathode loading or utilization below requirement	ED 483.25 Wh/kg vs 327.18 threshold (48 % margin); capacity 6.032 Ah; ceiling probe showed ED non-binding	Low	Low	Margin documented in DVPR item 2
6	Excessive SEI growth at 45 °C	High-temperature side reactions (ec reaction limited)	sei_thickness_nm_end 465.47 nm vs 550 nm (15 % margin)	Medium	Low	SEI kinetic rate at set value 1e-12 m/s; margin monitored; per-cycle capacity artifact labeled (not interpreted as real capacity)
7	Cooling degradation below design floor	Aging/damage of thermal interface lowering h	Sensitivity sweep: h=30 → TR at 666 s; h=25 → TR at 518.4 s	High	Medium (interface aging plausible in pack)	Design h = 50 with floor 40 (25 % margin); DVPR 6a is the evidence basis; pack-level verification required

Conclusion

Highest-risk items: #1 plating (mitigated: +0.048 V margin) and #2/#7 nail-TR/cooling degradation (mitigated: h = 50 vs floor ≈ 40). All mitigations are implemented in the current design (V11-final) and backed by simulation evidence; items #4 (molecular stability) remains an honest unassessed gap due to real_compute = false.